WEEK 10

REAL ESTATE ANALYTICS DASHBOARD

Introduction :

The Real Estate Analytics Dashboard was created using Power BI with the King County House Sales dataset. The main purpose of this dashboard is to analyse house prices, property sales, living areas, grades, conditions and locations through different visualisations.

The data preparation and transformation were completed in Week 9. In Week 10, the prepared dataset was used to create DAX measures and an interactive dashboard.

Objectives :

The main objectives of this project are:

1. To analyse house prices and property sales using Power BI.
2. To calculate important real estate measures using DAX.
3. To display key information using KPI cards.
4. To compare house prices based on bedrooms, bathrooms, waterfront, grade and condition.
5. To analyse property locations using Map and Filled Map visuals.
6. To use slicers for filtering properties by price and living area.
7. To create an interactive dashboard with three report pages.

DAX Measures Created :

Five DAX measures were created to calculate important real estate information.

1. Total Properties Sold :

Total Properties Sold = COUNTROWS(kc_house_data)

Counts the total number of property records.

2. Average House Price

Avg House Price = AVERAGE(kc_house_data[price])

Calculates the average house price.

3. Median Price

Median Price = MEDIAN(kc_house_data[price])

Calculates the middle value of house prices.

4. Average Living Area

Avg Living Area = AVERAGE(kc_house_data[sqft_living])

Calculates the average living area of houses.

5. Average Price per SqFt

Avg Price/SqFt = AVERAGE(kc_house_data[Price_per_SqFt])

Calculates the average price per square foot.

Real Estate Overview :

KPI Cards

Five Card visuals were created using the DAX measures:

1. Average Price per SqFt
2. Median Price
3. Total Properties Sold
4. Average Living Area
5. Average House Price

These cards display the important real estate details in a simple way.

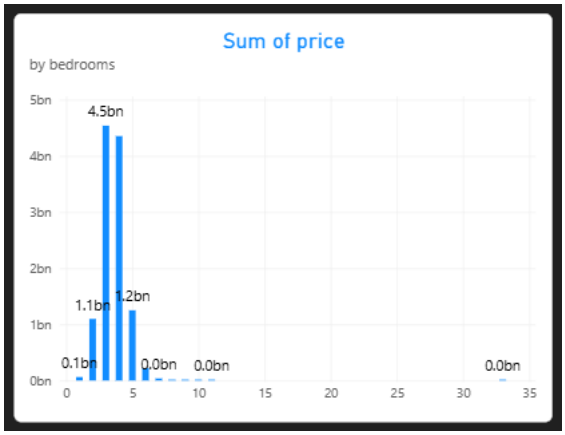


Charts

1. Clustered Column Chart – Sum of Price by Bedrooms

- X-axis: Bedrooms
- Y-axis: Sum of Price

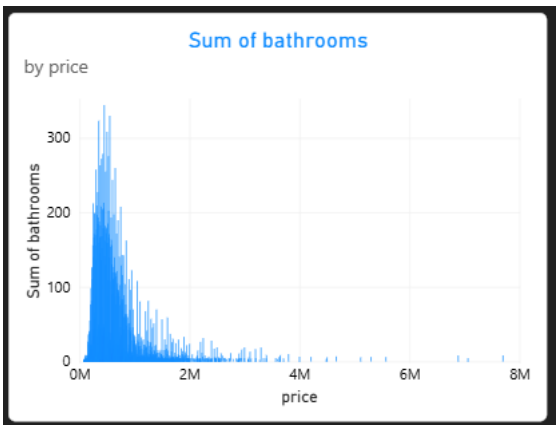
Shows the total house price based on the number of bedrooms.



2. Clustered Column Chart – Sum of Bathrooms by Price

- X-axis: Price
- Y-axis: Sum of Bathrooms

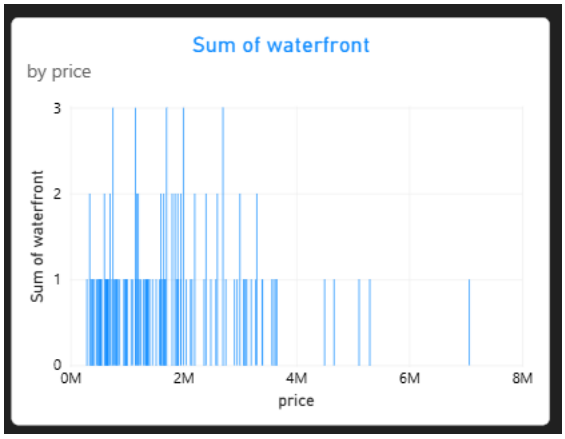
Shows the sum of bathrooms across different house price values.



3. Clustered Column Chart – Sum of Waterfront by Price

- X-axis: Price
- Y-axis: Sum of Waterfront

Shows waterfront values across different house price values.

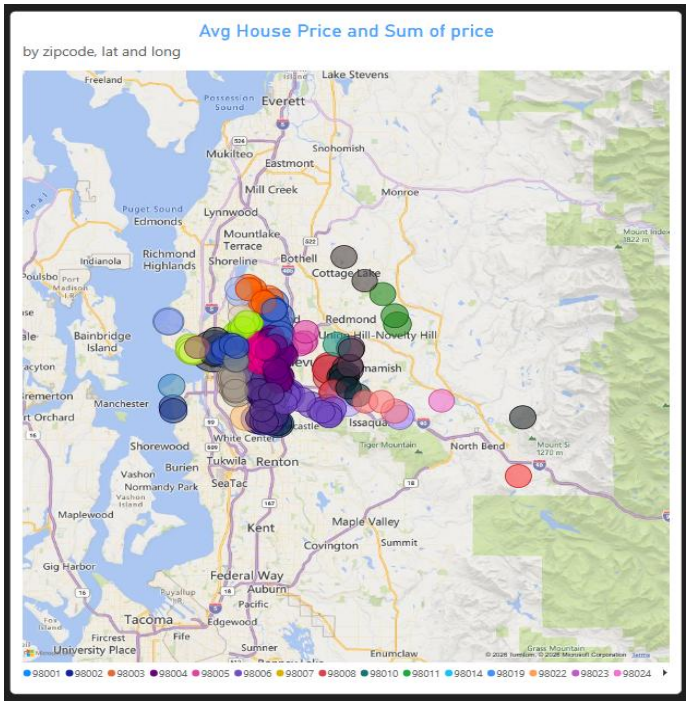


Property Location Analysis :

1. Map Visual – Average House Price and Sum of Price

- Legend: Zipcode
- Latitude: lat
- Longitude: long
- Bubble Size: Average House Price
- Tooltips: Sum of Price

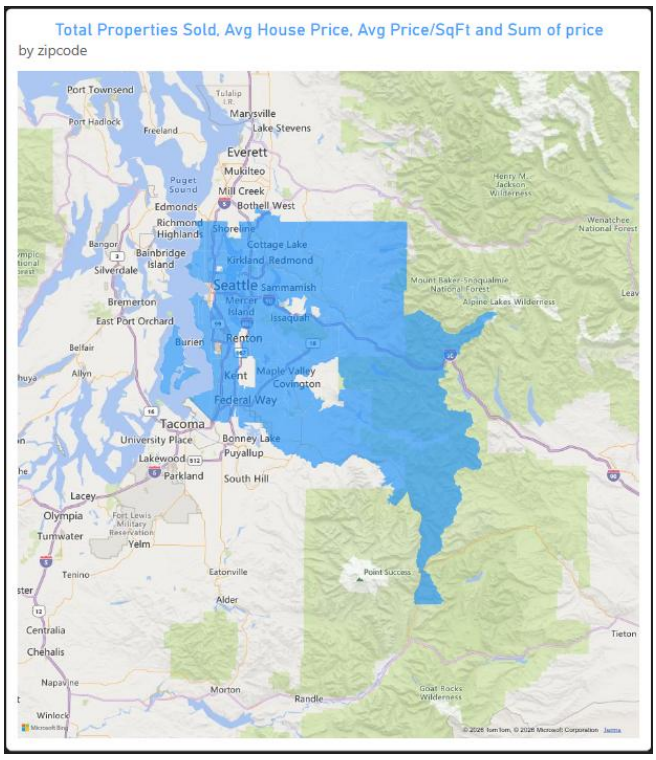
Displays property locations using map bubbles. It helps explore house prices in different geographical locations.



2. Filled Map – Property Details by ZIP Code

- Location: zipcode
- Tooltips: Total Properties Sold, Average House Price, Average Price per SqFt and Sum of Price

Displays property information across different ZIP code areas.

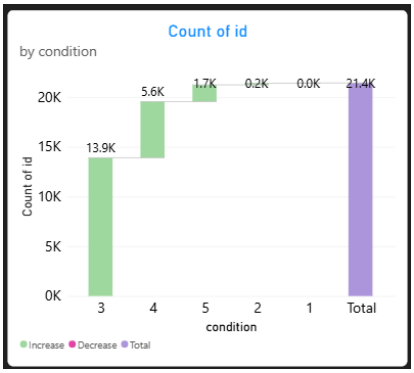


Detailed Property Analysis :

1. Waterfall Chart – Count of Properties by Condition

- Category: condition
- Y-axis: Count of id

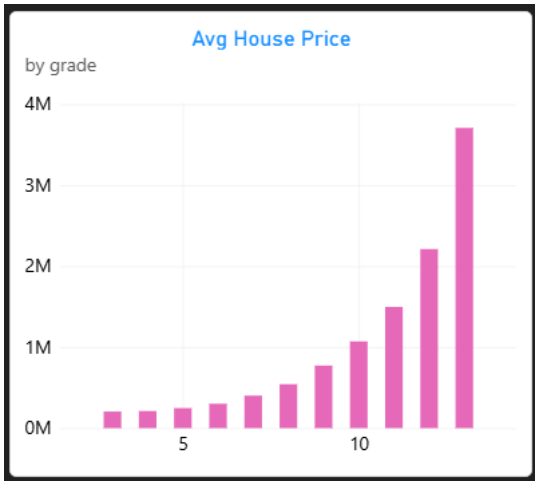
Shows the number of properties in each condition category.



2. Clustered Column Chart – Average House Price by Grade

- X-axis: grade
- Y-axis: Average House price

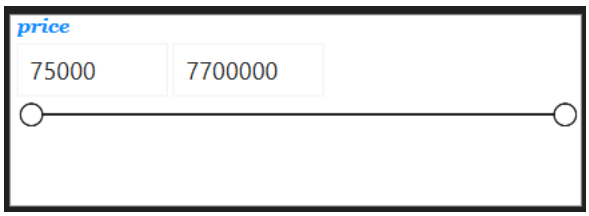
Compares average house prices across different property grades.



3. Price Range Slicer

- Field: price

Allows users to select a house price range and filter the dashboard.



4. Square Footage Slicer

- Field: sqft_living

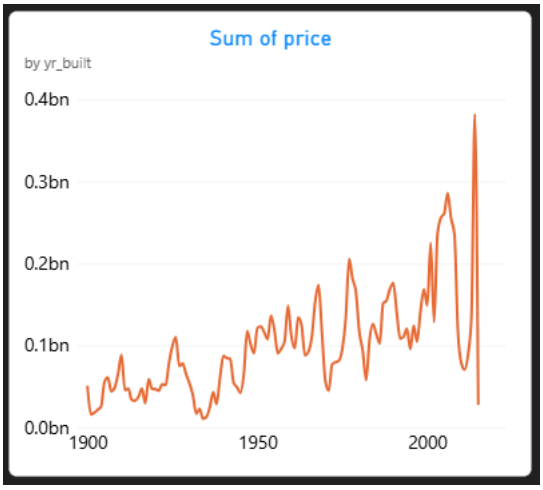
Allows users to filter properties based on their living area.



5. Line Chart – Sum of Price by Year Built

- X-axis: yr_built
- Y-axis: Sum of price

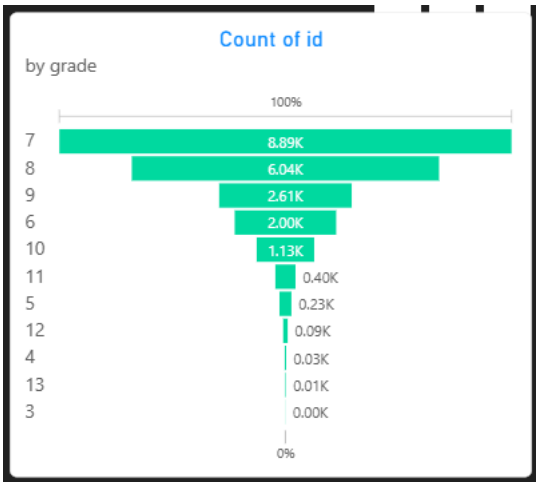
Shows the total house price based on the year the properties were built.



6. Funnel Chart – Count of Properties by Grade

- Category: grade
- Values: Count of id

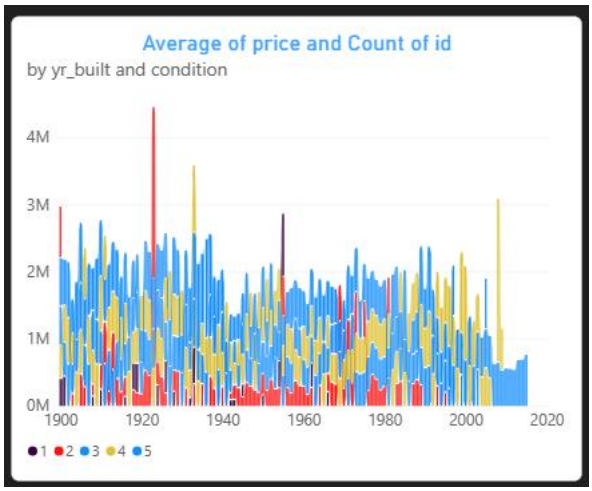
Shows the number of properties in each grade category.



7. Ribbon Chart – Average Price and Count of ID by Year Built and Condition

- X-axis: yr_built
- Y-axis: Average of price
- Tooltips: Count of id
- Legend: condition

Compares average house prices and property counts across construction years and conditions.



Interactive Features :

Two slicers were added to the dashboard:

- Price Range
- Square Footage

The slicers were configured using Sync Slicers to filter information across the report pages.

Conclusion :

The Real Estate Analytics Dashboard was successfully created using Power BI. Five DAX measures, KPI cards, different charts, maps and slicers were used to analyse the real estate dataset. The three dashboard pages provide a clear understanding of property prices, sales, conditions, grades and locations.